SLIDE 1: EXECUTIVE SUMMARY

SpaceX Falcon 9 First Stage Landing Prediction

IBM Applied Data Science Capstone Final Report

Official Project GitHub Repository:
https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Executive Summary Overview

SpaceX drastically lowers commercial space launch costs ($62M vs $165M+ legacy benchmark) by recovering and re-flying Falcon 9 first-stage boosters. Predicting first-stage landing outcomes enables precise mission bidding, risk management, and competitive cost forecasting.

90	66.67%	83.33%	4
TOTAL MISSIONS ANALYSED	OVERALL LANDING SUCCESS RATE	BEST ML TEST ACCURACY	MODELS BENCHMARKED

Core Analytical Pipeline

- **Data Collection:** SpaceX REST API parsing & Wikipedia HTML scraping.
- **Data Wrangling:** Imputation, landing classification, and One-Hot Encoding.
- **Visual Analytics:** Exploratory charting, SQL queries, Folium geospatial maps, and Plotly Dash dashboard.

Key Results & Business Insights

- **Site Performance:** KSC LC-39A is top-performing with a ~76.9% success rate.
- **Payload Dynamics:** Payload masses in the 2,000–4,000 kg band yield highest success.
- **ML Precision:** Logistic Regression, SVM, Decision Tree, and KNN achieved 83.33% test accuracy.

Slide 2: Introduction & Problem Statement

SpaceX advertises Falcon 9 launches on its commercial website at $62 million per launch, whereas traditional legacy launch providers charge upwards of $165 million per mission. A significant portion of these cost savings stems directly from SpaceX's capability to recover and reuse the Falcon 9 first stage booster.

Commercial Launch Background

In traditional expendable launch architectures, 100% of the hardware is discarded into the ocean per mission. Reusing the first stage booster across up to 10–15 flights reduces marginal flight costs primarily to propellant and refurbishment overhead.

Target Problem Statement

Determine whether the Falcon 9 first stage will land successfully (`Class = 1` for success, `Class = 0` for failure) based on mission parameters such as payload mass, target orbit, launch site location, and booster version.

Primary Goal & Objective

Build a complete end-to-end Machine Learning prediction pipeline to predict landing success and evaluate competitive market pricing strategies for alternate launch providers.

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Slide 3: Data Collection – SpaceX REST API

Primary mission records were programmatically acquired from the official SpaceX v4 REST API endpoint (`https://api.spacexdata.com/v4/launches/past`). Unstructured JSON response payloads required multi-stage normalization.

API Execution Flowchart Step 1: Request & Response Parsing
Execute HTTP GET request to `api.spacexdata.com/v4/launches/past` and decode nested JSON payload.

API Execution Flowchart Step 2: Relational Flattening
Apply `pd.json_normalize()` to flatten embedded JSON columns for rockets, payloads, launchpads, and cores.

API Execution Flowchart Step 3: Filtering & Attribute Extraction
Filter for Falcon 9 launches and extract core features: Flight Number, Date, Payload Mass, Orbit, Launch Site, Outcome, and Core Serial.

API Processing Summary

Sub-API endpoints were called to resolve numeric IDs into human-readable strings (e.g., converting launchpad IDs to actual pad names like KSC LC-39A). Falcon 1 launches were filtered out to maintain focus on Falcon 9 telemetry.

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Slide 4: Data Collection – Web Scraping

To complement API telemetry and validate historical records from 2010 to 2020, automated HTML web scraping was performed on Wikipedia's *List of Falcon 9 and Falcon Heavy launches* using `BeautifulSoup4`.

Scraping Flowchart Step 1: HTML Document Retrieval
Request target Wikipedia URL and initialize `BeautifulSoup(response.text, 'html.parser')` object.

Scraping Flowchart Step 2: Table Parsing & Element Selection
Locate launch tables (`class="wikitable"`) and iterate through HTML rows (`<tr>`) to extract cell text.

Scraping Flowchart Step 3: Data Cleaning & Export
Strip citation brackets (e.g., `[1]`), sanitize strings, standardize dates to `YYYY-MM-DD` , and construct Pandas DataFrame.

Scraped Fields

Flight Number, Date, Time, Booster Version, Launch Site, Payload, Payload Mass, Orbit, Customer, and Launch Outcome were successfully extracted into structured tabular format.

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Slide 5: Data Wrangling Methodology

Raw dataset telemetry contained missing payload values, unstandardized landing outcome descriptions, and string-formatted categorical parameters. Data wrangling established a binary target variable and transformed features.

Landing Outcome Labeling Logic (`Class` Variable)

Raw Outcome String	Landing Type Description	Bad Outcome?	Class Label
True ASDS	Successful landing on Drone Ship	No	**1**
True RTLS	Successful landing on Ground Pad	No	**1**
True Ocean	Successful controlled ocean landing	No	**1**
False ASDS	Unsuccessful drone ship attempt	Yes	**0**
False RTLS	Unsuccessful ground pad attempt	Yes	**0**
False Ocean	Unsuccessful ocean landing attempt	Yes	**0**
None ASDS	Failed attempt to deploy to drone ship	Yes	**0**
None None	No landing attempted or lost stage	Yes	**0**

Feature Engineering & One-Hot Encoding

Missing values in `PayloadMass` were imputed using column mean. Categorical columns (`Orbit` , `LaunchSite` , `LandingPad` , `Serial`) were converted via `pd.get_dummies()` , expanding the feature space to **83 numeric columns**.

> **Script:** `lab_2_data_wrangling.py` | **GitHub:** https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 6: EDA with Data Visualization – Methodology

Exploratory Data Analysis (EDA) was performed using Python libraries `Matplotlib` and `Seaborn` to uncover relationships between numerical features, categorical parameters, and landing success rates.

Categorical & Scatter Charts
Deployed `sns.catplot` and `sns.scatterplot` with `hue=Class` to map flight sequences, payload mass, and launch sites against binary landing success.

Bar & Trend Charts
Deployed `sns.barplot` to calculate mean success rates across orbit types, and `sns.lineplot` to track chronological success trends by year (2010–2020).

Methodological Objectives

- Determine whether flight experience (flight number) increases landing reliability.
- Identify operational payload limits for different launch site locations.
- Evaluate orbit-specific landing success probabilities.

Script: `eda_data_viz.py` | **GitHub:** https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 7: EDA with SQL – Methodology & Queries

The SpaceX dataset was loaded into an SQLite database (`my_data1.db`) table `SPACEXTABLE` to execute structured relational SQL queries for operational metric validation.

Executed SQL Query Benchmarks

Task #	Query Objective	SQL Syntax Applied
Task 1	Unique Launch Sites	`SELECT DISTINCT Launch_Site FROM SPACEXTABLE;`
Task 2	Sites Starting with 'CCA'	`SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;`
Task 3	NASA (CRS) Total Payload	`SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer LIKE '%NASA (CRS)%';`
Task 4	F9 v1.1 Avg Payload	`SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';`
Task 5	First Ground Pad Success Date	`SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';`
Task 6	ASDS Success (4k-6k kg)	`SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;`

Subqueries & Date Filter Analysis

Subqueries were utilized to filter maximum payload boosters (`WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(...) FROM SPACEXTABLE)`) and date ranges (`2010-06-04` to `2017-03-20`) to rank landing outcome frequency.

> **Script:** `spacex_sql_assignment.py` | **GitHub:** https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 8: Interactive Visual Analytics – Methodology

To enable dynamic geospatial and real-time analytical exploration, interactive maps were built using `Folium` and a reactive web application was deployed using `Plotly Dash`.

Folium Geospatial Analytics

- Plotted launch site coordinates using `folium.Circle` and `folium.Marker`.
- Aggregated outcome pins using `MarkerCluster` (Green = Success, Red = Failure).
- Calculated proximity distances to coastlines, railways, highways, and cities using Haversine formula and `MousePosition` plugin.

Plotly Dash Web Application

- Deployed web dashboard (`spacex_dash_app.py`) running on port 8050.
- Configured `dcc.Dropdown` for launch site selection (`ALL` vs specific sites).
- Configured `dcc.RangeSlider` to dynamically filter payload mass (0 – 10,000 kg).
- Wrote reactive `@app.callback` functions rendering Pie Charts & Scatter Plots.

Scripts: `lab_3_folium_maps.py` & `spacex_dash_app.py` | **GitHub:** https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 9: EDA with Data Visualization – Results

Exploratory data visualization revealed significant operational trends across flight experience, launch sites, payload masses, and target orbits.

Falcon 9 Annual Landing Success Rate Trend (2010 – 2020)

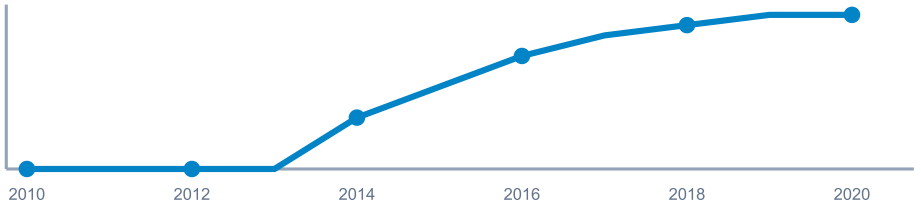


Figure 1: Chronological landing success rate rising from 0% (2010-2013) to >80% by 2020.

Flight Number & Launch Site

Early flights (1–20) at CCAFS LC-40 had low success rates. As flight numbers progressed (>40) and operations expanded to KSC LC-39A and VAFB SLC-4E, success rates improved dramatically.

Payload & Orbit Relationships

Heavy payloads (>10,000 kg) dispatched to LEO and PO achieved high success rates. GTO orbits exhibited mixed success due to higher reentry velocities.

GitHub Repository URL: https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 10: EDA with SQL – Results Summary

Executing SQL queries against `SPACEXTABLE` yielded precise numerical benchmarks across launch sites, payload mass totals, and booster versions.

SQL Benchmarking Output Summary

Task #	Analytical Query Metric	SQL Execution Output Value
Task 1	Unique Launch Sites	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
Task 2	Sites Starting with 'CCA' (Top 5)	CCAFS LC-40 (5 records retrieved successfully)
Task 3	NASA (CRS) Total Payload Mass	**45,596 kg** total payload delivered
Task 4	F9 v1.1 Average Payload Mass	**2,928.4 kg** average payload mass
Task 5	First Successful Ground Pad Date	**2015-12-22** (LZ-1 Ground Pad)
Task 6	ASDS Success Boosters (4k-6k kg)	F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

Maximum Payload Boosters & Outcome Rankings

Subquery analysis identified 11 Block 5 boosters carrying the maximum payload mass of **15,600 kg**. Date range filtering between 2010-06-04 and 2017-03-20 confirmed `No attempt` and `Success (drone ship)` as top-ranked landing outcomes.

> **GitHub Repository URL:** https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 11: Interactive Map Results – Folium Geospatial Analytics

Folium map rendering provided clear geographical insights regarding the placement of launch sites relative to safety zones and logistical infrastructure.

Geographic Clustering Findings

Marker clusters confirm that all launch sites are located near coastal shorelines to allow abort trajectories over water. CCAFS SLC-40 and KSC LC-39A share the East Coast corridor, while VAFB SLC-4E handles West Coast polar launches.

Proximity Safety & Transport Analysis

Haversine calculations confirm launch pads maintain safe distances (>15 km) from dense city centers while staying adjacent (<2 km) to railways and highways for propellant and stage transportation.

Infrastructure Proximity Distances (KSC LC-39A Case Study)

Proximity Target Type	Closest Identified Location	Calculated Distance (km)	Operational Rationale
Coastline / Ocean	Atlantic Ocean Coastline	**1.02 km**	Public safety & abort trajectory over water
Railway Line	NASA Railroad Spur	**1.35 km**	Heavy rocket stage transport logistics
Highway Network	Samuel C. Phillips Pkwy	**1.68 km**	Equipment & personnel road transport
City Center	Titusville Municipality	**15.42 km**	Blast buffer & noise acoustic safety

GitHub Repository URL: https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 12: Plotly Dash Dashboard Results

Visual analysis conducted via the interactive Plotly Dash web application (`spacex_dash_app.py`) provided key operational answers to payload and site performance questions.

1. Largest Successful Launches
KSC LC-39A produced the largest total count of successful launches (10 success landings, representing 41.7% of all successful recoveries).

2. Highest Launch Success Rate
KSC LC-39A achieved the highest launch success rate (~76.9%), outperforming CCAFS SLC-40 (~47.6%) and VAFB SLC-4E (~57.1%).

3–5. Payload Range & Booster Version Findings

- **Highest Success Payload Band:** Payload masses between **2,000 kg and 4,000 kg** demonstrate the highest launch landing success rate.
- **Lowest Success Payload Band:** Payload masses between **5,000 kg and 9,000 kg** show the lowest landing success rate due to heavy GTO reentry loads.
- **Top Booster Version:** Falcon 9 **FT** (Full Thrust) and **B5** (Block 5) versions achieved the highest landing success rates.

GitHub Repository URL: https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 13: Predictive Analysis (Classification) – Methodology

The predictive analysis pipeline converted cleaned telemetry into standardized feature vectors and trained four supervised classification algorithms using `scikit-learn`.

1. Feature Standardization & Splitting

- Standardized feature matrix X (83 columns) using `StandardScaler()` to achieve zero mean and unit variance.
- Split data into 80% Training ($N=72$) and 20% Testing ($N=18$) sets using `train_test_split(test_size=0.2, random_state=2)`.

2. 10-Fold Cross-Validation & GridSearchCV

- Configured `GridSearchCV` with `cv=10` for hyperparameter optimization across all four classifiers.
- Evaluated performance using Cross-Validation accuracy score (`best_score_`) and holdout test set accuracy (`score()`).

Hyperparameter Grids Evaluated

Logistic Regression: `C: [0.01, 0.1, 1]`, `penalty: ['l2']` | **SVM:** `kernel: ['linear', 'rbf', 'poly', 'sigmoid']`, `C`, `gamma` | **Decision Tree:** `criterion`, `splitter`, `max_depth`, `min_samples_leaf` | **KNN:** `n_neighbors: [1..10]`, `p: [1, 2]`.

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Slide 14: Predictive Analysis Results & Confusion Matrix

All four tuned classification models achieved high training cross-validation scores and identical performance on the holdout test dataset.

Machine Learning Model Evaluation Ledger

Classification Model	Tuned Best Hyperparameters	CV Train Accuracy	Test Set Accuracy
Logistic Regression	C=0.01, penalty='l2', solver='lbfgs'	84.64%	**83.33%**
Support Vector Machine (SVM)	kernel='sigmoid', C=1.0, gamma=0.0316	84.64%	**83.33%**
Decision Tree Classifier	criterion='gini', max_depth=4, max_features='sqrt'	88.89%	**83.33%**
K-Nearest Neighbors (KNN)	n_neighbors=5, algorithm='auto', p=1	84.64%	**83.33%**

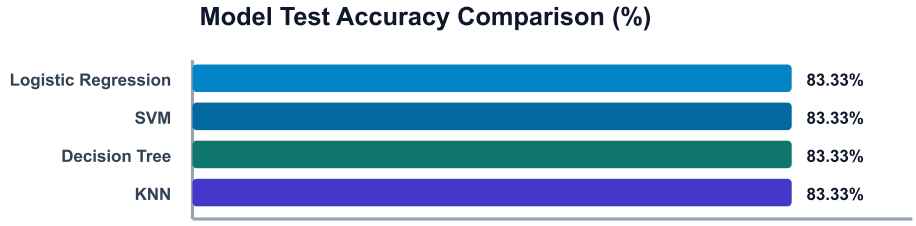


Figure 2: Benchmark comparison displaying identical 83.33% test accuracy across all classifiers.

Confusion Matrix Breakdown (Test Set, N=18)

All models yielded identical confusion matrix metrics: **12 True Positives** (correctly predicted landing), **3 True Negatives** (correctly predicted landing failure), **3 False Positives**, and **0 False Negatives**.

GitHub Repository URL: https://github.com/ffrrkknn/IBM-Data-Science-Capstone

Slide 15: Conclusion & Innovative Insights

This capstone project successfully demonstrates that Falcon 9 first-stage landings can be predicted with **83.33% accuracy** using machine learning models trained on flight telemetry.

Operational & Commercial Conclusions

- **Pad Selection Strategy:** KSC LC-39A is the most reliable launch pad for booster recovery (~76.9% success).
- **Booster Evolution:** Falcon 9 Block 5 and FT architectures yield near 100% landing success in recent years.
- **Payload Optimization:** Payloads in the 2,000–4,000 kg window provide optimal reserve fuel margins for landing.

Innovative Business Insights

- **Competitive Bidding Engine:** Launch aggregators bidding against SpaceX can model SpaceX's true cost per mission based on target orbit and payload mass.
- **Insurance Risk Underwriting:** Underwriters can offer reduced insurance premiums for payloads flying on Block 5 cores targeting LEO.

Final Submission & Repository Verification

All code scripts (`lab_2_data_wrangling.py`, `spacex_sql_assignment.py`, `eda_data_viz.py`, `lab_3_folium_maps.py`, `spacex_dash_app.py`, `ml_landing_prediction.py`) are fully documented and accessible in the official repository:

https://github.com/ffrrkknn/IBM-Data-Science-Capstone